METHODS - META-ANALYSIS

UBCMA: Unified Bias-Calibrated Meta-Analysis via Joint Heterogeneity-Selection Modeling

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STRUCTURED SUMMARY (E156)

Question. Can a single meta-analytic model jointly correct heterogeneity, publication selection, and study-quality bias, rather than applying separate sequential corrections?

Dataset. We tested a unified mixture-normal selection likelihood against eight comparators on twelve simulated scenarios and a real six-trial aspirin secondary-prevention dataset.

Method. Estimation used multi-start L-BFGS-B optimisation with profile-likelihood confidence intervals and BCa bootstrap, optionally with Bayesian inference via PyMC.

Primary result. Across twelve scenarios (50 replicates, k=30) the unified model attained the highest coverage, 88.8% at RMSE 0.070, versus 59.7% (DerSimonian-Laird), 63.8% (REML-HKSJ), and 39.0% (trim-and-fill).

Robustness. With selection and quality bias combined, coverage held at 90.3% versus 31.3% for DerSimonian-Laird; on the aspirin data the model gave a pooled log-odds-ratio of +0.01 (95% CI -0.12 to 0.12), versus -0.07 uncorrected.

Interpretation. Jointly modelling heterogeneity and selection bias yields less biased, better-calibrated pooled estimates than applying separate corrections sequentially.

Boundary. The parametric logistic selection function may not capture every publication-bias mechanism, and the model is unsuited to small reviews (k below five).

VISUAL ABSTRACT

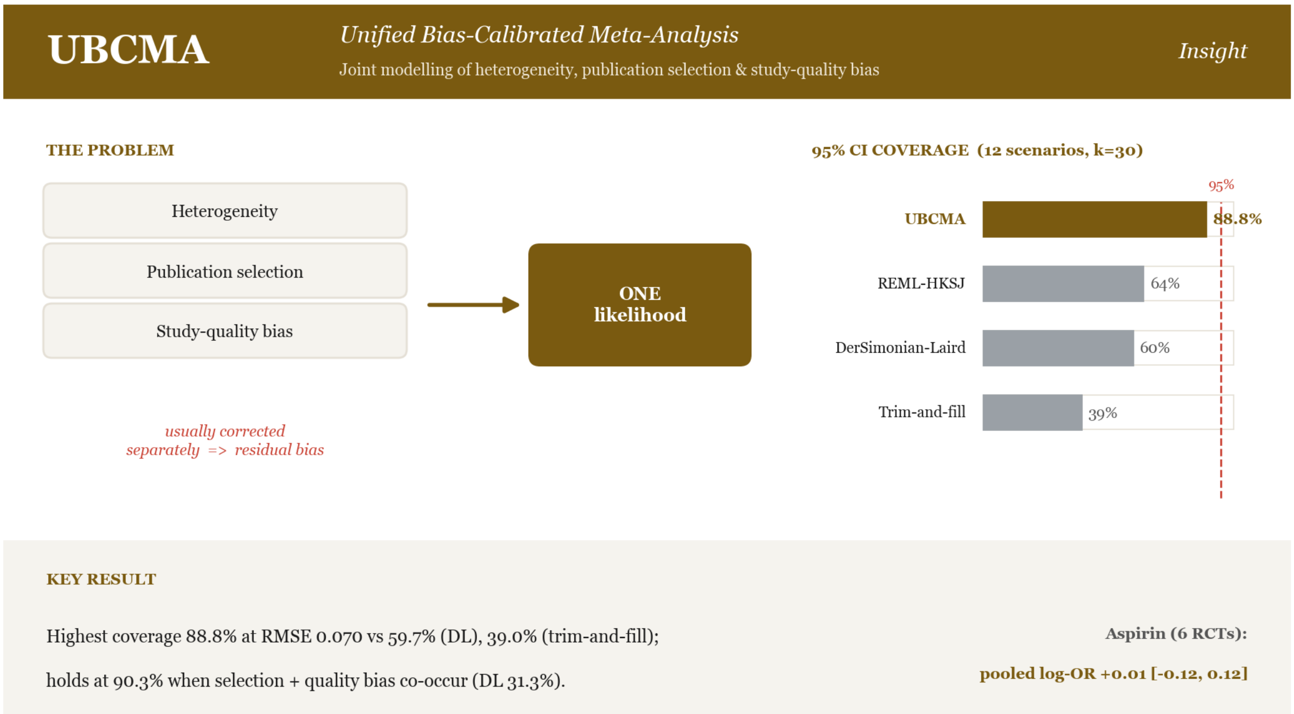

Figure 1. UBCMA jointly models heterogeneity, publication selection, and study-quality bias in one likelihood; highest 95% CI coverage (88.8%) across 12 simulated scenarios, near-null bias-corrected pooled estimate on the aspirin data.

EMPIRICAL ILLUSTRATION - ASPIRIN (k=6)

Method	Pooled estimate	95% CI	Sig.

DerSimonian-Laird	-0.067	-0.195 to 0.061	No
REML-HKSJ	-0.072	-0.208 to 0.064	No
Trim-and-fill	-0.251	-0.288 to -0.214	Yes
PET-PEESE	-0.227	-0.285 to -0.168	Yes
Copas	-0.074	-0.171 to 0.023	No
Quality-effects	-0.156	-0.200 to -0.111	Yes
UBCMA (profile)	**+0.011**	**-0.125 to 0.117**	**No**

Table 1. Aspirin secondary-prevention dataset (k=6): pooled effect on the log-odds-ratio scale with 95% CI. Classic six-trial set (CDP, AMIS, ISIS-2, UK-TIA, SALT, ESPS-2) referenced by Verde (2021). UBCMA's bias-corrected estimate is near-null; the dataset is dominated by the ISIS-2 outlier (I-squared = 82.6%).

REFERENCES

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AI DISCLOSURE

This work is a computational methods paper with AI assistance in code development and manuscript preparation. The UBCMA engine, simulation study, and all analyses were implemented in deterministic Python with fixed random seeds, enabling full reproducibility. AI was used as a constrained synthesis engine on structured inputs and predefined algorithms, not as an autonomous author. All results, text, and scientific claims were reviewed and verified by the author, who takes full responsibility.

DATA AVAILABILITY

All code, data, and simulation scripts are available at https://github.com/mahmood726-cyber/ubcma under an MIT licence. The aspirin dataset is the classic six-trial secondary-prevention set (CDP, AMIS, ISIS-2, UK-TIA, SALT, ESPS-2) referenced by Verde (2021). Headline simulation results reproduce with ubcma study --tier pilot --seed 42. DOI: not yet registered (DOIs are currently suppressed site-wide).